

Name:

Roll No.:

Make Sure Attempt All Questions (No choice is available).

Total Marks: 110

Time: 150 minutes

Q1. Write the Outputs of following codes.

Marks: 60

```
#include <iostream>
using namespace std;
int main()
{
    int x= 5;
    int y= 5;

    {
        int x= 5;
        x= 6;
    }

    cout << x << endl;
    cout << y << endl;

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int x= 5;
    int y= 5;

    {
        x= 6;
        y= 6;
    }

    cout << x << endl;
    cout << y << endl;

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int num = 10;

    if(num == 10);
    {
        cout << "Number is Correct" << endl;
    }
    else
    {
        cout << "Incorrect Number" << endl;
    }

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int num;

    if (!(num >20 && num <30 ))
    {
        cout << "Number is Correct" << endl;
    }
    else
    {
        cout << "Incorrect Number" << endl;
    }

    return 0;
}
// Write the output for the numbers 15 and 25.
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{

    char var1 = 'A';

    char var2 = 65;

    char var3 = '9';

    cout << var1 << endl;
    cout << var2 << endl;
    cout << var3 << endl;

    return 0;

}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int num = 10;

    for (int i=2 ; i <= num ; i++)
    {
        if(num % i == 0)
        {
            continue;
        }

        cout << i << endl;

    }

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int num = 15;
    bool flag = true;
    int c1 = 2;
    int c2;

    while(c1 <= num)
    {
        c2 = 2;
        while(c2 < c1)
        {
            if(c1 % c2 == 0)
            {
                flag = false;
                break;
            }
            c2++;
        }

        if(flag)
        {
            cout<< c1 << endl;
        }
        else{
            flag = true ;
        }
        c1++;
    }

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int size = 5;
    int arr [size]= {1,2,3,4,5};

    cout << arr << endl;
    cout << arr[size] << endl;
    cout << sizeof(arr) << endl;

    for(int i=1 ; i<=size ;i++)
    {
        cout << arr[i-1] << endl;
    }

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{

    int a = 10;
    int b = a++;
    int c = ++b;
    int d = ++a + c ;

    cout << a + b + c + d << endl;

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{

    char arr[] ={'h', 'e', 'l', 'l', 'o','\0'};
    cout << arr << endl;
    cout << arr + 2 << endl;
    cout << sizeof(arr) << endl;

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int temp = 0 ;
    int S=0;
    int key = 5;
    int arr [] = {1 , 5 , 3, 4, 5, 6, 7 , -1};

    while(arr[S]!= -1)
    {
        if(arr[S] > 0)
        {
            S++;
        }
    }

    for (int i=0; i<S ; i++)
    {
        if(arr[i]== key)
        {
            arr[i]= (key * 2);
        }
    }

    for (int i=0; i<S ; i++)
    {
        cout << arr[i] << " ";
    }

    return 0;
}
```

Output, Errors and dry Run.

Name:

Roll No.:

```
#include <iostream>
using namespace std;
int main()
{
    int num = 10 ;
    for(int i=1 ; i <= num ; i+=2 )
    {

        cout<< i << endl;

    }

    return 0;
}
```

How many times that this loop execute if num = 15

no of times (header) =

no of times (body) =

Output, Errors and dry Run.

Name:

Roll No.:

Subjective Type (Use of Compiler)

Q2. Write the following C++ programs.

Marks: 50

1. C++ program to find greater number between two numbers without using relational operator. (>, <, >=, <=) and conditional statement (if, else, switch). 5
2. C++ Program to take a number from user and print complete table of that number without using Multiply Operation. (*). 5
3. C++ program to convert Decimal to Binary number system. 5
4. C++ program to take a number from user and determine if number is the *Fibonacci sequence* then output Yes it is fibo else No it is not fibo. 5
5. Print the following pattern from user input. 8

Enter number of Rows = 4	Enter number of Rows = 5
1	1
101	1 1
10101	1 2 1
1010101	1 3 3 1
	1 4 6 4 1

6. C++ program to find Kth smallest and largest Number in Array. 7
7. C++ program to take two numbers form user if 1st number have 3 times and 2nd number have 2 times same number then output Yes, otherwise Output No 10

Example:

Input	Input	Input
num1 = 1122333555; num2 = 133222444;	num1 = 11223555; num2 = 133222444;	num1 = 1120003555; num2 = 102220444;
3 is 3 time in num1 3 is 2 time in num2	5 is 3 time in num1 But 5 is not 2 time in num2	0 is 3 time in num1 0 is 2 time in num2
Output Yes	Output No	Output Yes

8. C++ program to take an arrays from user and find median, mean and mode of Sorted Array. 5